

STRUCTURED NOMINAL AND MODAL REFERENCE

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This dissertation investigates the notion of *discourse reference* (Karttunen 1976) involved in the interpretation of anaphora and quantification in natural language. The main proposal is that both nominal and modal expressions introduce (possibly) non-singleton *sets of objects* (individuals and possible worlds respectively) and that these sets are *correlated* in discourse: discourse reference involves two equally important components with essentially the same interpretive dynamics, namely reference to *values*, i.e. the sets of objects, and reference to *structure*, i.e. the correlation / dependency between such sets, which is introduced and incrementally elaborated upon in discourse.

To define and investigate structured discourse reference, a new dynamic system couched in classical (many-sorted) type logic is introduced which extends Compositional DRT (CDRT, Muskens 1996). Given the underlying type logic, compositionality at the sub-clausal level follows automatically and standard techniques from Montague semantics (e.g. type shifting) become available.

The main technical innovation relative to CDRT is that, just as in Dynamic Plural Logic (van den Berg 1996), information states are modeled as *sets* of variable assignments; such *plural* info states can be represented as matrices with assignments (sequences) as rows.

Plural info states encode discourse reference to both values and structure. The values are the sets of objects that are stored in the *columns* of the matrix, e.g. a discourse referent (dref) *u* for individuals stores a set of individuals relative to a plural info state, since *u* is assigned an individual by each assignment (i.e. row). The structure is *distributively* encoded in the *rows* of the matrix: for each assignment / row in the plural info state, the individual assigned to a dref *u* by that assignment is structurally correlated with the individual assigned to some other dref *u'* by the same assignment. The resulting system is dubbed Plural CDRT.

In Plural CDRT (PCDRT), sentences denote *relations* between an input and an output plural info state. Indefinites non-deterministically introduce both values and structure, i.e. they

introduce *structured sets* of individuals; pronouns are anaphoric to such structured sets. Quantification is defined in terms of matrices instead of single assignments and the semantics of the non-quantificational part becomes rules for how to fill out a matrix.

PCDRT enables us to give a compositional account of the *mixed weak & strong* donkey sentence in (1) below and of the *donkey anaphora to structure* exemplified by sentence (2).

1. Every u_1 person who buys a u_2 book on [amazon.com](https://www.amazon.com) and has a u_3 credit card uses it u_3 to pay for it u_2 .
2. Every u_1 boy who bought a u_2 Christmas gift for a u_3 girl in his class asked her u_3 deskmate to wrap it u_2 .

Moreover, the fact that we work with finer-grained meanings (given our plural info states) enables us to analyze complex interactions between *individual-level* and *modal* anaphora by simply adding dref's p, p' etc. for possible worlds: such a dref p stores a set of worlds, i.e. a *proposition*, relative to a plural info state and it can be *structurally* correlated with some other modal dref p' and / or with individual-level dref's u, u' etc.

The resulting Intensional PCDRT system captures the truth-conditions of and the modal and individual-level anaphoric connections established in naturally occurring discourses like (3) below. The interaction between modalized conditionals, donkey anaphora, modal subordination and the entailment particle *therefore* is more transparent in the equivalent discourse in (4) below.

3. a. [A] man cannot live without joy. b. Therefore, when he is deprived of true spiritual joys, it is necessary that he become addicted to carnal pleasures. (Thomas Aquinas, attributed)
4. a. If P_1 a u_1 man is alive, he u_1 must p_1 find something u_2 pleasurable / he u_1 must p_1 have a u_2 pleasure.
b. Therefore, if p_1 P_2 he u_1 doesn't have any u_3 spiritual pleasure, he u_1 must p_2 have a u_4 carnal pleasure.

The present investigation is located at the intersection of two research programs. On the one hand, it takes the program in Muskens (1996) of unifying Montague semantics and dynamic semantics one step further: Intensional PCDRT unifies – in classical type logic – the static Lewis (1973) / Kratzer (1981) analysis of modal quantification and van den Berg's Dynamic Plural Logic. On the other hand, Intensional PCDRT enables us to explicitly and systematically capture the anaphoric and quantificational parallels between the individual and modal domains (e.g. individual-level and modal dref's have parallel types), thus bringing further support to the conjecture that our semantic competence is domain neutral, first put forth in Partee (1973) and further pursued in Stone (1999), Bittner (2001) and Schlenker (2005) among others.

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